Edoardo Mangia

Education

Chalmers University of Technology — Gothenburg, Sweden *(Sept. 2024 - ...in progress)*
M.Sc. Engineering Mathematics (GPA 4.0/4.0)
High-Performance Computing (4.0/4.0) - Bayesian Machine Learning (4.0/4.0) - Compiler Construction (?/4.0)

University of Padua — Padua, Italy *(Sept. 2020 - Jan. 2024)*
B.Sc. Industrial Engineering (GPA 3.6/4.0)
Introductory Computer Science (4.0/4.0) - Operations Research (4.0/4.0) - Electrotechnics (4.0/4.0)

Experience

CEA – Commissariat à l'énergie atomique — Saint-Paul-lès-Durance, France *(Jun. 2026 - Nov. 2026)*
Research Intern
I'll join the IRESNE institute at the Cadarache site, working on high-performance simulation tools for nuclear fuel modeling. The project focuses on GPU-acceleration of the phase-field finite-element solver `SLOTH`, part of the `PLEIADES` simulation platform used for nucluear fuel behavior studies.

Jülich Research Centre — Jülich, Germany *(Feb. 2026 - June 2026)*
Research Intern
In the IBG-1 group, I was contributing to their open-source `hopsy` and `PolyRound` repositories. Specifically, I was working on improving polytopes simplification and rounding algorithms, both touching the logic and by making them multithreadead and GPU-accelerated.

MAX IV Laboratory — Lund, Sweden *(Nov. 2025 - Feb. 2026)*
Research Intern
The project was conducted in collaboration between my university and the MAX IV Laboratory, a synchrotron light source facility. The focus was on building a simulation model to understand how X-rays interact with specific materials in different settings, reducing sample damage in synchrotron experiments (XCT, SWAXS, XPS). The model computes dosage prediction, radiolysis and thermal effects, using libraries from Geant4 and gVXR.

ASML — Veldhoven, Netherlands *(Feb. 2025 - Jul. 2025)*
Research Intern
I was in the reticle-frontside R&D team, addressing nanoparticles contamination on EUV scanners reticles. More in detail, this involved numerically modeling the effects of the EUV-plasma, electrostatics, chemical processes in the scanner. I was also developing a custom algorithm to identify clusters of particles on the reticle, based on their material composition, size and transient particle bursts.

Skills

Programming Languages (Python, C, C++, Julia, Rust)
Parallel Computing (CUDA, MPI, OpenMP)
Machine Learning (PyTorch, NumPy, scikit-learn)
Linux and Bash Scripting (SLURM, vim, gdb)
Languages English (IELTS C1), French, Spanish, Russian (basic conversational level).

Projects (GitHub)

Building a compiler in C++ for Javalette, a C-like language, with LLVM backend targeting ARM and RISC-V.
Developing a GPU-accelerated Poisson solver in CUDA C++.
Reimplemented research papers of personal interest and some other university coursework.